SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	**192572101**

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I M.Kyathi Sai Priya (192572101) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _M.Kyathi___

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

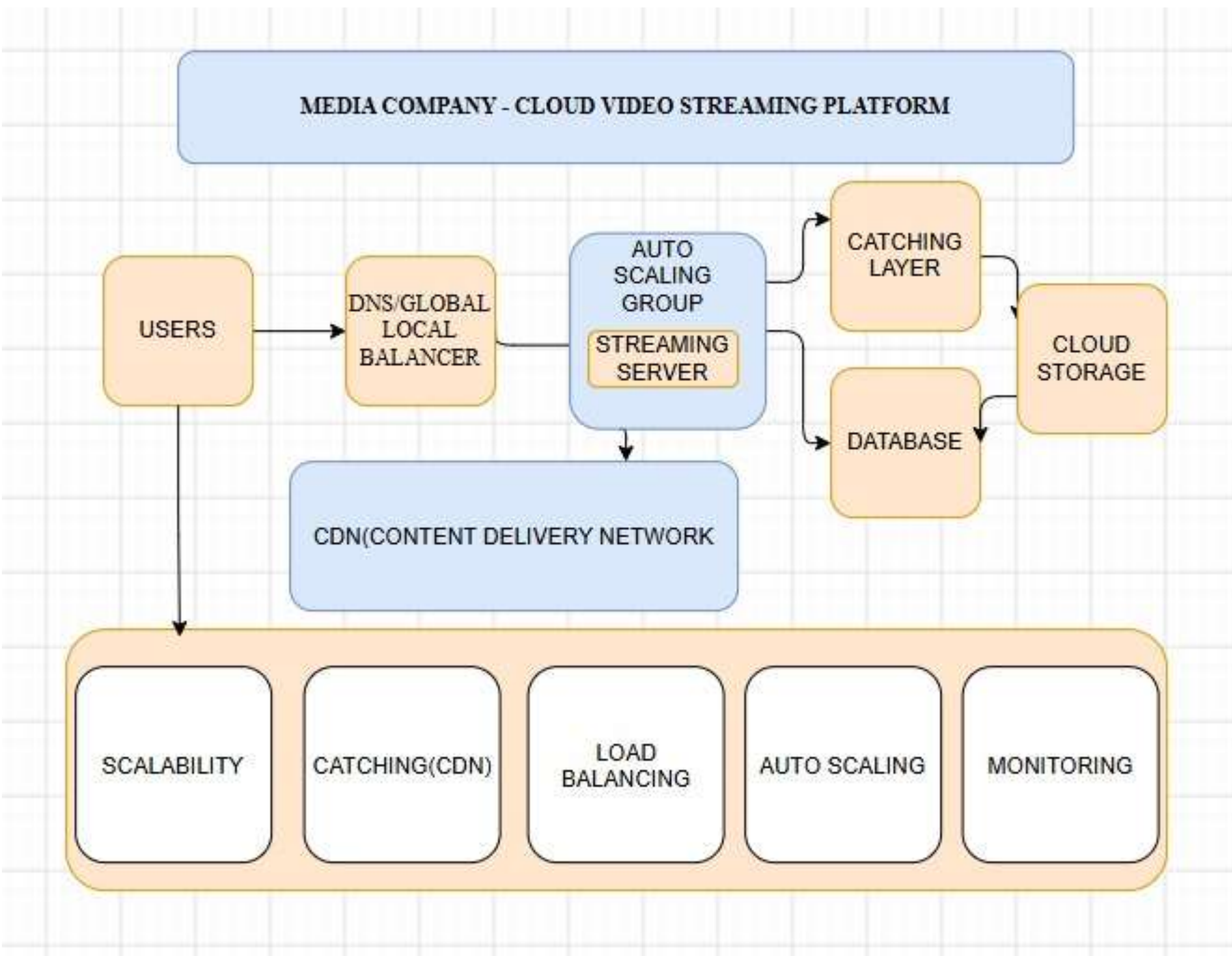
Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 – Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Ans: Media Company – Cloud-Based Video Streaming Platform :

Cloud Architecture :

- Store videos in cloud object storage.
- Use a Content Delivery Network (CDN) to deliver videos from servers close to users.
- Deploy multiple application servers behind a Load Balancer**.**
- Use Auto Scaling to automatically add or remove servers based on user traffic.
- Use Caching (Redis/Memcached) to store frequently accessed content and reduce database load.



How Load Balancing and Auto Scaling Ensure Smooth Streaming

Load Balancing:
Load balancing distributes incoming user requests evenly across multiple application servers. Instead of sending all users to a single server, it directs each request to the server with the least workload or the best availability. This prevents any one server from becoming overloaded, reduces response time, and ensures that users experience smooth video playback with minimal buffering. If one server fails, the load balancer automatically redirects traffic to other healthy servers, maintaining uninterrupted service.

Auto Scaling:
Auto scaling automatically adjusts the number of application servers based on user demand. During peak hours, when millions of users access the streaming platform, new server instances are launched automatically to handle the increased traffic. When demand decreases, unnecessary servers are removed to reduce cloud costs. This ensures that sufficient computing resources are always available, providing consistent streaming performance while optimizing resource usage.

Working Together:
Load balancing and auto scaling work together to provide a seamless streaming experience. The load balancer evenly distributes user requests among all available servers, while auto scaling ensures that enough servers are available to handle changing traffic levels. As a result, the platform can support millions of concurrent users, minimize buffering, maintain high availability, and optimize operational costs.

Cost Optimization Strategies :

- Use Auto Scaling to pay only for required resources.
- Store old or less-used videos in cheaper storage tiers.
- Use CDN caching to reduce bandwidth costs.
- Use Reserved or Spot Instances for predictable workloads.
- Continuously monitor resource usage and remove unused resources.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies.

Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Ans : Fintech Startup – Multi-Cloud Container Deployment

Proposing a Solution Using Container Orchestration and Multi-Cloud

- Package applications using **Docker** containers.
- Manage containers using **Kubernetes** across different cloud providers.
- Store container images in a centralized container registry.
- Use Infrastructure as Code (IaC) for consistent deployment.

Explaining that how Service Discovery and Scaling Can Be Handled Efficiently :

Service discovery:

Service discovery helps different containers and microservices find and communicate with each other automatically. In a Kubernetes cluster, every service is assigned a unique DNS name and IP address. When a new container starts or an existing one stops, Kubernetes automatically updates the service information. This eliminates the need for manual configuration and ensures reliable communication between application components.

Scaling:

Scaling is managed automatically using Kubernetes. When the application experiences increased traffic or higher CPU and memory usage, Kubernetes creates additional container instances (pods) to handle the load. When the demand decreases, it removes the extra instances to save resources and reduce costs. A built-in load balancer distributes incoming requests evenly among all running containers, ensuring efficient resource utilization and consistent application performance.

Overall:

By combining automatic service discovery with auto-scaling, the application remains highly available, performs efficiently under varying workloads, reduces manual management, and provides a seamless experience for users across multiple cloud providers.

Benefits

- High availability.

- Avoids dependence on one cloud provider.
- Better disaster recovery.
- Improved flexibility.
- Optimized costs by choosing suitable cloud services.

Risks

- Higher management complexity.
- Increased networking challenges.
- Different cloud services may not be fully compatible.
- Security management becomes more difficult.
- Higher operational cost.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Ans :

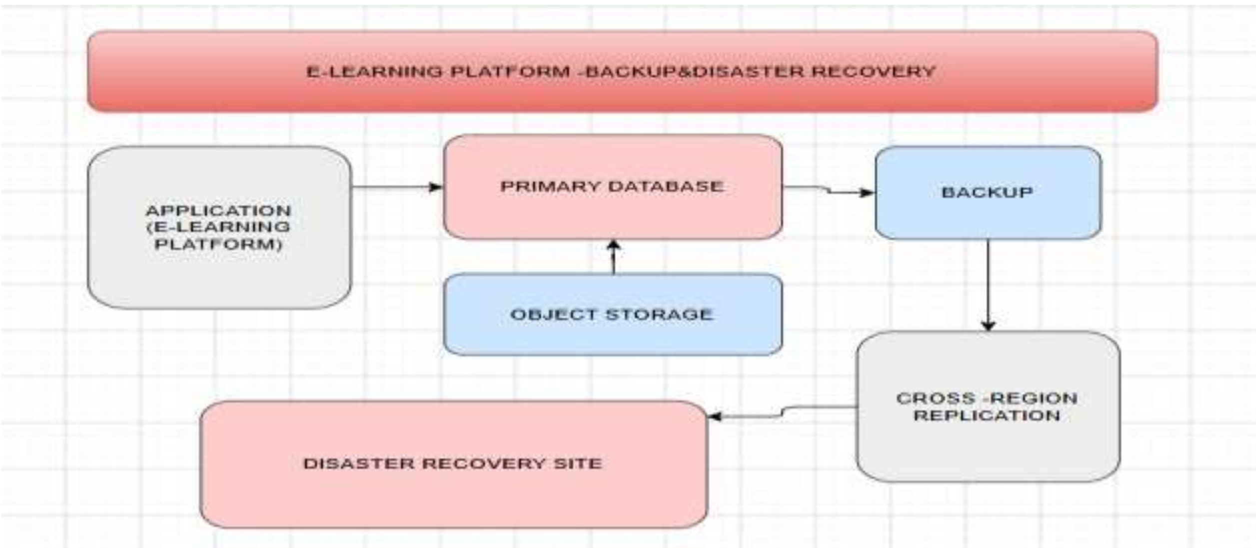
Designing a Solution Using Cloud-Native Backup, Replication, and Versioning

To prevent data loss and ensure quick recovery, the e-learning platform should use a combination of cloud-native backup, replication, and versioning**.**

- **Cloud-Native Backup:** Configure automatic backups of databases, student records, course materials, and application data at regular intervals. These backups are securely stored in cloud storage and can be restored whenever needed.
- **Replication:** Enable continuous replication of data from the primary cloud region to a secondary region. If the primary region becomes unavailable due to a disaster or outage, the replicated copy can be used to restore services with minimal downtime.

- **Versioning:** Enable versioning for cloud storage so that every time a file is modified or deleted, the previous versions are retained. This allows administrators to recover accidentally deleted or overwritten files without affecting other data.

Architecture



Explaining how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) Are Achieved

Recovery Time Objective (RTO) :

RTO is the maximum acceptable time required to restore services after a failure.

It is achieved by:

- Maintaining a standby environment in another cloud region.
- Using automated backup restoration.
- Configuring automatic failover to the replicated system.
- Regularly testing disaster recovery procedures.

As a result, the application can resume operation within the target recovery time.

Recovery Point Objective (RPO)

RPO is the maximum amount of data that can be lost during a failure.

It is achieved by:

- Performing frequent automatic backups.
- Continuously replicating data to another region.
- Enabling object versioning for important files.
- Synchronizing databases in near real time.

This ensures that only a very small amount of recent data, if any, is lost during a disaster.

Steps to Ensure Business Continuity During Failures

1. Schedule automatic backups at regular intervals.
2. Enable versioning to recover deleted or modified files.
3. Replicate critical data to a secondary cloud region.
4. Configure automatic failover to the backup environment.
5. Continuously monitor backup and replication status.
6. Perform regular disaster recovery testing to verify backup integrity.
7. Maintain clear recovery procedures and train administrators.
8. Monitor system health and send alerts for failures or backup issues.
9. Periodically review and update the disaster recovery plan.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Ans :

Government Agency – Hybrid Cloud Architecture :

Design a Hybrid Architecture that Ensures Secure Communication Between Environments :

In a hybrid cloud, sensitive government data is kept in the on-premises data center to meet security and compliance requirements, while non-sensitive applications and public services are hosted in the public cloud. Both environments are connected through a secure VPN or dedicated private connection to ensure safe communication and data transfer

Explanation:

- Sensitive information such as citizen records and confidential government applications remains in the on-premises data center.
- Public-facing websites, portals, and less-sensitive applications are hosted in the cloud.
- A secure VPN or private network connection encrypts communication between the two environments.
- Centralized monitoring ensures both environments are managed efficiently and securely.

Explain How Identity Management and Encryption Should Be Implemented

Identity Management

Identity management ensures that only authorized users can access cloud resources.

It can be implemented by:

- Using **Single Sign-On (SSO)** so users log in once to access both on-premises and cloud applications.
- Enabling **Multi-Factor Authentication (MFA)** to provide an extra layer of security.
- Applying **Role-Based Access Control (RBAC)** so users receive permissions based on their job roles.
- Managing user identities through a centralized identity provider to maintain consistent access control across both environments.

Encryption

Encryption protects data from unauthorized access.

It should be implemented by:

- Encrypting **data at rest** (stored data) using strong encryption algorithms.
- Encrypting **data in transit** using secure protocols such as TLS/SSL while transferring data between the cloud and on-premises systems.
- Storing and managing encryption keys securely using a centralized key management service.
- Regularly rotating encryption keys to improve security.

Challenges in Managing Hybrid Cloud Environments

- Managing resources across both on-premises and cloud environments increases operational complexity.
- Maintaining consistent security policies and access controls can be difficult.
- Ensuring compliance with government regulations requires continuous monitoring and auditing.
- Network latency between cloud and on-premises systems may affect application performance.
- Integrating different platforms and technologies can be challenging.
- Monitoring, troubleshooting, and managing multiple environments require skilled personnel and advanced management tools.
- Higher maintenance and operational costs compared to using a single environment.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

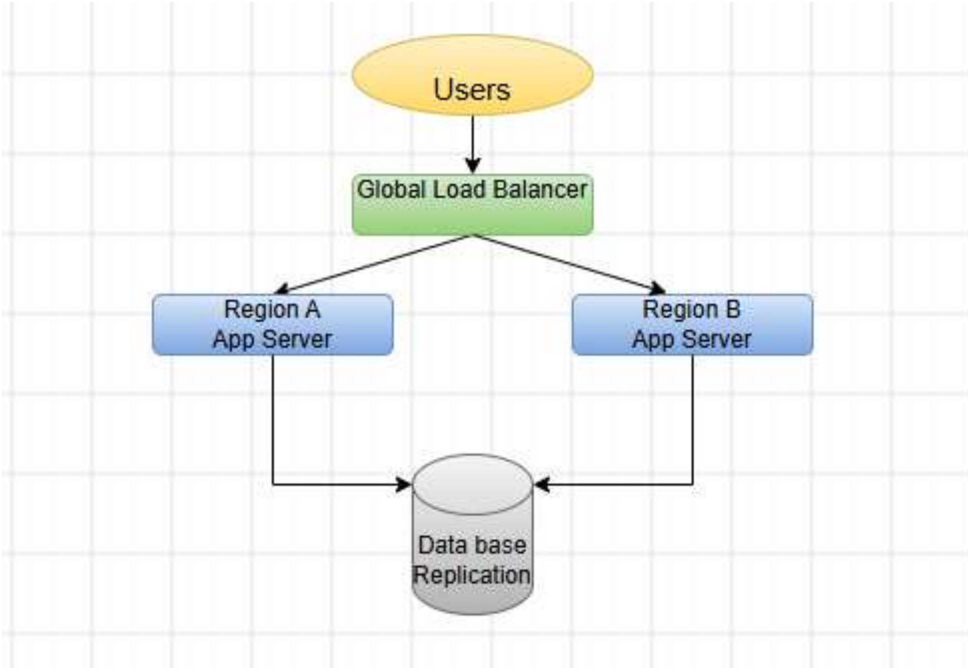
Ans :

SaaS Provider – Multi-Region Cloud Deployment

1. Proposng a Multi-Region Deployment Strategy

To ensure high availability and fault tolerance, the SaaS application should be deployed across multiple cloud regions**.** Each region hosts identical application servers and databases so that users can continue accessing the application even if one region experiences a failure.

A **Global Load Balancer** distributes user requests to the nearest or healthiest region based on availability and performance. If one region becomes unavailable, traffic is automatically redirected to another healthy region without interrupting the service.



Explanation:

- Deploy the same application in two or more cloud regions.
- Use a Global Load Balancer to route user requests to the nearest available region.
- Synchronize application data across all regions.
- If one region fails, users are automatically redirected to another healthy region.

Explaining how Data Replication and Failover Mechanisms Should Be Configured :

Data Replication

Data replication ensures that the same data is available in multiple regions.

It can be configured by:

- Continuously replicating databases from the primary region to secondary regions.
- Synchronizing application files and storage between regions.
- Performing regular backups to protect against accidental data loss.
- Using synchronous or asynchronous replication depending on business requirements.

Failover Mechanism

Failover ensures continuous service during failures.

It can be configured by:

- Continuously monitoring the health of application servers and databases.
- Automatically switching user traffic to a healthy region if the primary region becomes unavailable.
- Using DNS or Global Load Balancer failover to redirect requests.
- Regularly testing the failover process to ensure it works correctly during real failures.

This configuration minimizes downtime and maintains uninterrupted service for users.

- Discussing a monitoring and alerting Strategies for Proactive Issue Detection
- Effective monitoring helps identify problems before they affect users.

The following strategies can be implemented:

- Continuously monitor CPU, memory, disk usage, network traffic, and application performance.
- Perform regular health checks on servers, databases, and services.
- Use centralized logging to collect and analyze logs from all cloud regions.
- Configure alerts for high CPU usage, low storage space, server failures, unusual network activity, and application errors.
- Send notifications through email, SMS, or messaging platforms so administrators can respond quickly.
- Create dashboards to monitor system performance and resource utilization in real time.
- Regularly review monitoring reports to identify performance trends and prevent future issues.